Sovereign Spreads and the Snowball Effect: Panel Evidence from the Euro Area Periphery

Country Risk Analysis — Research Paper

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Abstract

This paper investigates the relationship between sovereign spreads and the snowball effect of public debt dynamics—the interest–growth differential $r - g$—across the three largest peripheral euro area economies: Italy, Portugal, and Spain (IPS), over the period 1996–2019. Extending a single-country Italian analysis to a balanced panel with country fixed effects and Driscoll–Kraay standard errors, we establish three main findings. First, the sovereign spread is the dominant driver of the snowball effect across all IPS countries: a one-percentage-point increase in the spread is associated with a 0.83 pp increase in $r - g$ in the baseline fixed-effects specification, surviving multiple robustness checks including two-way fixed effects and mean-group estimation. Second, we identify a common transmission threshold of 1.23 percentage points—consistent across countries and identical to the threshold estimated for Italy alone—above which the marginal effect of the spread on the snowball effect diminishes rather than amplifies, contradicting models of explosive debt spirals. Third, the post-2010 attenuation of this channel is less statistically pronounced in the panel than in single-country Italian regressions, suggesting that ECB backstop effects were heterogeneous: strongest for Italy, which benefited directly from the "whatever it takes" commitment, and weaker for Portugal and Spain, which were subject to formal bailout conditionality. We further show that adding Greece to the sample collapses threshold identification—the estimated threshold jumps to 5.71 pp (90th percentile), capturing almost exclusively Greek crisis observations—and reverses the post-2010 sign, confirming that the Greek episode is structurally distinct and contaminates inference for the IPS group.

Keywords: Sovereign spreads; snowball effect; debt sustainability; fiscal panel; ECB; threshold effects; structural break; euro area periphery.

JEL: F34, G15, H63, E43, C23.

1 Introduction

The sustainability of public debt in a monetary union without a common fiscal backstop depends critically on the feedback between market sentiment and debt dynamics. When investors demand higher risk premia—reflected in wider sovereign yield spreads—the effective interest rate on government debt rises. If this increase outpaces nominal economic growth, the interest–growth differential $r - g$, known in fiscal analysis as the *snowball effect*, turns positive, causing debt-to-GDP to accumulate mechanically even in the absence of primary deficits. This channel is particularly consequential in high-debt economies where the stock of outstanding liabilities amplifies the sensitivity of debt dynamics to rate movements.

The euro area sovereign debt crisis of 2010–2012 provided a dramatic natural experiment in this transmission mechanism. Peripheral sovereigns experienced abrupt repricing of sovereign risk as markets unwound the convergence trade that had prevailed since 1999. Spreads over the German Bund widened by hundreds of basis points within months, tightening financial conditions, depressing growth prospects, and worsening the snowball component of debt accumulation simultaneously. The eventual stabilization, triggered by the ECB's commitment to do "whatever it takes" in July 2012 and the announcement of the Outright Monetary Transactions (OMT) programme, illustrated that the same mechanism could operate in reverse: credible institutional backstops could compress spreads, reduce financing costs, and interrupt self-fulfilling debt spirals (De Grauwe and Ji, 2013, 2014).

Existing empirical work on the spread–snowball nexus has focused predominantly on individual countries or on the determinants of spreads themselves (Corsetti et al., 2014; Zoli, 2013). A single-country time-series approach is informative about the case under study but cannot address whether the same structural parameters hold for other peripheral economies, whether threshold effects in the spread–snowball relationship are country-specific, or whether the post-2010 attenuation reflects a country-specific policy response or a broader eurozone-wide regime change. Panel data, even with a small cross-section, allow us to exploit within-country variation while controlling for country-specific structural differences and to test parameter stability across countries.

This paper fills this gap. Using a balanced panel of Italy, Portugal, and Spain (IPS) over 1996–2019, we estimate a sequence of models of increasing complexity: a baseline country fixed-effects specification, a post-2010 structural break model, a threshold model that endogenously identifies the kink in the spread–snowball relationship, and a full model combining both. We complement this with four robustness exercises and an extension that adds Greece to the sample—explicitly to assess how a structurally outlying crisis episode distorts inference on the common parameters.

Our main contributions are the following.

- We establish that sovereign spreads are the dominant, robust determinant of the snowball effect across the IPS periphery, with an average elasticity of 0.83 pp per pp in the baseline specification and 1.27 pp per pp in the pre-2010 period.
- We identify a common transmission threshold of $k^* = 1.23$ pp—the same value found for Italy in isolation—providing strong cross-country validation of the nonlinear mechanism.
- We document that the post-2010 attenuation of the spread–snowball channel is less uniform in the panel than in the Italian time series, consistent with heterogeneous ECB backstop exposure across peripheral countries.
- We show that including Greece collapses threshold identification and reverses the sign of the post-2010 interaction, confirming that the Greek crisis is a structural outlier that should be separated from IPS inference.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 describes the data and estimation strategy. Section 4 presents the IPS panel results. Section 5 reports robustness checks. Section 6 presents the Greek extension. Section 7 interprets the findings. Section 8 discusses limitations and Section 9 concludes.

2 Related Literature

Debt sustainability and the snowball effect. The decomposition of changes in the debt-to-GDP ratio into the primary balance, the snowball component, and stock-flow adjustments is a cornerstone of fiscal analysis (Blanchard, 2019; Bohn, 1998). The snowball effect, $\mathrm{sb}_t = r_t - g_t$, determines whether debt accumulates mechanically even under a balanced primary budget. Blanchard (2019) formalises the condition $r < g$ as the boundary between dynamically efficient and inefficient debt regimes, arguing that persistently low real interest rates can make high debt ratios sustainable without immediate fiscal adjustment. Euro area evidence documented by European Central Bank (2009) shows that episodes of debt reduction in Belgium and Finland during the 1990s were driven by favourable interest–growth differentials, while Italy's persistently adverse $r - g$ contributed to debt accumulation despite moderate primary surpluses.

Sovereign spreads and the risk channel. De Grauwe and Ji (2013) show that euro area spreads were significantly driven by self-fulfilling market sentiment rather than fundamentals alone, particularly in the 2010–2012 crisis period. Corsetti et al. (2014) formalise the sovereign risk channel: an increase in the sovereign risk premium raises financing costs for both the government and the private sector, depressing investment and growth, which in turn worsens fiscal sustainability. This feedback loop can generate multiple equilibria—a good equilibrium with low spreads and stable debt dynamics, and a bad equilibrium with high spreads and deteriorating dynamics. Zoli (2013) provides empirical estimates of spread pass-through to bank funding costs and lending rates in Italy, confirming the relevance of the channel in the Italian case. Balduzzi et al. (2020) document how political uncertainty and populist episodes can raise spreads independently of fiscal fundamentals, adding an additional layer of sovereign risk transmission.

ECB interventions and regime change. The ECB's evolving role as an implicit or explicit backstop for sovereign debt markets constitutes the primary source of structural change in the spread–debt dynamics nexus after 2010. De Grauwe and Ji (2014) show that the OMT announcement caused a significant and permanent reduction in the sensitivity of peripheral spreads to self-fulfilling sentiment, effectively breaking the doom loop between default fears and borrowing costs. Pietrunti and Zaghini (2020) provide direct empirical evidence that OMT reduced sovereign risk premia and improved monetary policy transmission in Italy. Corsetti et al. (2015) examine the broader institutional reforms of the post-crisis period and their implications for debt sustainability. Bacchiocchi and Borghi (2016) document the specific role of ECB unconventional monetary policy in containing Italian sovereign spreads.

Panel econometric methods for small-N fiscal panels. Driscoll and Kraay (1998) propose the nonparametric covariance estimator—the Driscoll–Kraay estimator—that is consistent in the presence of general forms of cross-sectional and temporal dependence, suitable for the small-N, large-T panels typical of comparative fiscal studies. Im et al. (2003) develop the panel unit root test (IPS $\bar{t}$ statistic) that pools country-level ADF tests. Pesaran and Smith (1995) propose the mean-group (MG) estimator that allows slope heterogeneity across units; we use it as a robustness check. Hansen (1999) develops the threshold panel regression framework that we adapt to identify the common kink in the spread–snowball relationship.

3 Data and Empirical Framework

3.1 Sample and Variables

The baseline sample covers Italy (ITA), Portugal (PRT), and Spain (ESP) over 1996–2019, yielding a balanced panel of $N \times T = 3 \times 24 = 72$ observations. The sample is bounded below by 1996 because consistent AMECO series begin in that year, and above by 2019 to exclude the COVID-19 shock, which generates snowball dynamics driven by an exceptional GDP collapse rather than movements in sovereign risk premia.

The **snowball effect** is defined as

$$\mathrm{sb}_{it} = r_{it} - g_{it}, \tag{1}$$

where r_{it} is the implicit interest rate on general government debt (interest expenditure divided by the lagged stock of debt) and g_{it} is nominal GDP growth, both in percentage points. A positive snowball indicates that debt grows mechanically even with a zero primary balance; a negative value indicates the reverse.

The **sovereign spread** (spr_{it}) is the yield on the 10-year government benchmark bond over the 10-year German Bund, sourced from the ECB Statistical Data Warehouse (ECB SDW). We use the spread rather than the yield level because it isolates the country-specific risk premium from the common euro area term structure.

Control variables are drawn from AMECO and include: the lagged **debt-to-GDP ratio** (b_{it-1}, % of GDP); the **fiscal stance** (fs_{it}), measured as the cyclically adjusted primary balance (% GDP; more negative = looser stance); real **GDP growth** (g_{it}, %); and the **inflation rate** (π_{it}, GDP deflator, %).

3.2 Descriptive Statistics

Table 1 summarises the pooled and country-level statistics for the IPS sample. Italy displays the highest average snowball (2.44 pp) and debt-to-GDP ratio (118%), reflecting its structural vulnerability. Portugal shows the greatest dispersion in both the snowball (sd = 2.94) and the sovereign spread (sd = 2.36), driven by the dramatic entry into and exit from a financial assistance programme during 2011–2014. Spain exhibits the lowest average snowball (0.21 pp) and debt ratio (69%), consistent with its pre-crisis fiscal starting position. The pooled average snowball of 1.13 pp alongside an average spread of 1.37 pp indicates that peripheral euro area

Table 1: Descriptive Statistics: IPS Baseline Panel, 1996–2019

	Snowball	Spread	Debt/GDP	Fiscal stance	Growth	Inflation
Pooled ($N = 72$)						
Mean	1.133	1.372	92.07	−3.436	1.399	2.069
Std	2.575	1.653	30.91	2.094	2.242	1.285
Min	−2.513	0.020	35.70	−11.19	−5.305	−1.933
Max	10.091	9.050	134.8	0.167	5.201	4.149
Italy ($N = 24$)						
Mean	2.441	1.261	118.1	−2.786	0.600	1.922
Std	2.170	1.109	11.79	1.606	1.878	1.113
Portugal ($N = 24$)						
Mean	0.744	1.816	89.52	−4.453	1.435	2.174
Std	2.941	2.364	30.63	2.163	2.290	1.398
Spain ($N = 24$)						
Mean	0.213	1.038	68.58	−3.069	2.161	2.112
Std	2.069	1.130	24.13	2.147	2.341	1.368

Sources: AMECO (snowball, debt/GDP, fiscal stance, growth, inflation); ECB SDW (sovereign spread over German Bund, 10-year maturity). Fiscal stance = cyclically adjusted primary balance (% GDP). Snowball = $r - g$ in percentage points.

countries faced persistently adverse interest–growth differentials accompanied by non-trivial sovereign risk premia over the full sample.

3.3 Model Specifications

We estimate five nested specifications. Throughout, i indexes country, t indexes year, α_i denotes country fixed effects, and ε_{it} is the error term.

Baseline (M2): Country fixed-effects model.

$$\text{sb}_{it} = \alpha_i + \beta_1 \text{spr}_{it} + \beta_2 b_{it-1} + \beta_3 \text{fs}_{it} + \beta_4 g_{it} + \beta_5 \pi_{it} + \varepsilon_{it}. \quad (2)$$

Country fixed effects absorb permanent cross-country differences in the level of $r - g$. Italy serves as the reference category.

Post-2010 break (M3).

$$\text{sb}_{it} = \alpha_i + \beta_1 \text{spr}_{it} + \beta_2(\text{spr}_{it} \times D_{it}^{2010}) + \beta_3 D_{it}^{2010} + \beta_4 b_{it-1} + \beta_5 \text{fs}_{it} + \beta_6 g_{it} + \beta_7 \pi_{it} + \varepsilon_{it}, \quad (3)$$

where $D_{it}^{2010} = \mathbf{1}(t \geq 2010)$. The marginal effect of the spread is β_1 before 2010 and $\beta_1 + \beta_2$ after 2010. A negative β_2 indicates attenuation.

Threshold model (M4). Following Hansen (1999), we endogenously estimate a common threshold k^* by minimising the sum of squared residuals (SSR) over a grid of candidate values drawn from the interior of the spread distribution:

$$\mathrm{sb}_{it} = \alpha_i + \beta_1 \mathrm{spr}_{it} + \beta_2\, \mathrm{spr}_{it} \cdot \mathbf{1}(\mathrm{spr}_{it} > k^*) + \beta_3 b_{it-1} + \beta_4 \mathrm{fs}_{it} + \beta_5 g_{it} + \beta_6 \pi_{it} + \varepsilon_{it}. \tag{4}$$

The marginal effect is β_1 below the threshold and $\beta_1 + \beta_2$ above it.

Full model (M5). The full specification combines the temporal break and the threshold nonlinearity:

$$\begin{aligned} \mathrm{sb}_{it} = \alpha_i &+ \beta_1 \mathrm{spr}_{it} + \beta_2\, \mathrm{spr}_{it} \cdot \mathbf{1}(\mathrm{spr}_{it} > k^*) + \beta_3\, (\mathrm{spr}_{it} \times D_{it}^{2010}) \\ &+ \beta_4\, [\mathrm{spr}_{it} \cdot \mathbf{1}(\mathrm{spr}_{it} > k^*) \times D_{it}^{2010}] + \beta_5 D_{it}^{2010} + \boldsymbol{\gamma}' \mathbf{Z}_{it} + \varepsilon_{it}, \end{aligned} \tag{5}$$

where $\mathbf{Z}_{it} = (b_{it-1}, \mathrm{fs}_{it}, g_{it}, \pi_{it})'$. This specification yields four regime-specific marginal effects of the spread on the snowball.

3.4 Inference Strategy

All standard errors are Driscoll–Kraay (DK) (Driscoll and Kraay, 1998), which are consistent for balanced panels with $T \to \infty$ in the presence of cross-sectional dependence, heteroskedasticity, and serial correlation of unknown form. The DK variance estimator uses a Newey–West kernel with bandwidth m:

$$\hat{\Omega}_{\mathrm{DK}} = \sum_{s=-m}^{m} \left(1 - \frac{|s|}{m+1}\right) \hat{\Gamma}(s), \tag{6}$$

where $\hat{\Gamma}(s) = T^{-1} \sum_t \hat{h}_t \hat{h}_{t-s}'$ and $\hat{h}_t = \sum_i x_{it} \hat{u}_{it}$. We set $m = 2$ (approximately $T^{1/4}$ for $T = 24$). Panel unit root properties are discussed in Appendix B; in brief, flow-rate variables (snowball, spread, growth, inflation, fiscal stance) are treated as $I(0)$ and the debt-to-GDP ratio as near-$I(1)$.

4 Baseline Results: The IPS Panel

4.1 Fixed-Effects Baseline

Column (1) of Table 2 reports the baseline fixed-effects model (M2). The sovereign spread enters with a coefficient of 0.830 ($\mathrm{SE}_{\mathrm{DK}} = 0.097$, $t_{\mathrm{DK}} = 8.54$, $p < 0.001$), indicating that a one-percentage-point increase in the spread is associated with a 0.83 pp increase in the snowball effect. The coefficient is economically significant: the spread's standard deviation of 1.65 pp implies that a one-standard-deviation increase is associated with a 1.37 pp increase in $r - g$, equivalent to 53% of the snowball's own standard deviation.

The lagged debt ratio enters negatively ($\hat{\beta}_2 = -0.036$, $t_{\mathrm{DK}} = -4.15$, $p < 0.001$). This counterintuitive sign reflects a pattern documented for Italy: in peripheral euro area economies, high debt levels have historically coexisted with accommodative monetary policy and compressed effective rates rather than automatically triggering market discipline through higher spreads. The fiscal stance is statistically insignificant throughout, consistent with the view that $r - g$

in the periphery is driven primarily by market conditions and ECB policy rather than by the primary balance position alone.

Real GDP growth has the expected negative sign ($\hat{\beta}_4 = -0.584$, $t_{\mathrm{DK}} = -5.78$, $p < 0.001$): stronger growth mechanically compresses $r - g$. Inflation also enters negatively ($\hat{\beta}_5 = -0.277$, $t_{\mathrm{DK}} = -3.50$, $p = 0.002$), consistent with higher nominal growth compressing the differential when the effective nominal interest rate adjusts more sluggishly than prices.

The country fixed effects are large and precisely estimated: Portugal's average snowball is 2.63 pp lower than Italy's, and Spain's is 2.90 pp lower, both significant at the 1% level. These differences reflect Italy's structurally higher effective interest burden given its outstanding debt stock above 100% of GDP throughout the sample, versus Spain's much lower starting debt ratio and Portugal's intermediate position. The model R^2 rises to 0.867 from 0.788 in the pooled OLS benchmark.

4.2 Post-2010 Structural Break

Column (2) of Table 2 reports model (M3). Before 2010, the estimated marginal effect of the spread on the snowball is 1.270 pp per pp ($t_{\mathrm{DK}} = 8.61$, $p < 0.001$). The interaction term spr $\times$ D^{2010} carries a coefficient of -0.354 ($t_{\mathrm{DK}} = -1.66$, $p = 0.111$), implying a post-2010 marginal effect of $1.270 - 0.354 = 0.916$ pp per pp.

The direction of this attenuation is consistent with the single-country Italian evidence and with the hypothesis that ECB interventions compressed the pass-through from spreads to effective borrowing costs. However, the interaction is statistically insignificant at the 5% DK level in the panel, in contrast to the highly significant attenuation found in the Italian time series. This discrepancy is substantively informative rather than a failure of the panel specification.

Portugal entered a formal EU-IMF assistance programme in 2011 and exited in 2014 under strict conditionality, while Spain received a more targeted bank recapitalisation facility. Neither country benefited in the same way as Italy from the direct sovereign debt market commitment that OMT represented: the "whatever it takes" announcement was explicitly directed at containing self-fulfilling spread dynamics for countries retaining market access, and Italy was its primary beneficiary. The pooled post-2010 interaction therefore averages an attenuation that was strong for Italy and weaker for the two countries under formal assistance conditionality, diluting the panel estimate toward statistical insignificance. The heterogeneity of the ECB backstop effect thus emerges as a genuine empirical finding, not an artefact of the panel structure.

4.3 Threshold Model

Grid search over the interior of the spread distribution identifies an optimal threshold $k^* = 1.23$ pp at the 60th percentile of the pooled spread distribution. This estimate is strikingly consistent with the threshold estimated for Italy alone, providing strong cross-country validation. Column (3) of Table 2 shows that below the threshold the spread coefficient is 1.683 ($t_{\mathrm{DK}} = 2.50$, $p = 0.020$), while the high-spread increment is -0.783 ($t_{\mathrm{DK}} = -1.34$, $p = 0.193$). The implied marginal effect above the threshold is $1.683 - 0.783 = 0.900$ pp per pp.

The negative sign on the high-spread increment contradicts models of convex amplification in which sovereign stress becomes self-reinforcing above a critical level (Corsetti et al., 2014).

Instead, the point estimates consistently suggest that the transmission of spreads to debt dynamics *flattens* at high spread levels. A natural interpretation is that episodes of very high spreads also tend to be episodes in which ECB intervention was most likely or in which political pressure for fiscal adjustment was strongest, limiting the mechanical pass-through from market rates to actual borrowing costs even before any formal announcement. The threshold of 1.23 pp corresponds approximately to the spread level below which investor discrimination among euro area sovereigns was negligible prior to the GFC—a regime of effectively pooled sovereign risk. Above it, markets began treating sovereign risk as country-specific, triggering a qualitative change in how spread movements translate into financing conditions.

4.4 Full Model: Four-Regime Marginal Effects

The full model (M5), reported in column (4) of Table 2, decomposes the marginal effect of the spread into four regimes. Figure 1 visualises these estimates. The regime-specific marginal effects are:

- **Pre-2010, low spread** (spr $\leq$ 1.23 pp): +2.303
- **Pre-2010, high spread** (spr $>$ 1.23 pp): +1.267
- **Post-2010, low spread** (spr $\leq$ 1.23 pp): +0.692
- **Post-2010, high spread** (spr $>$ 1.23 pp): +0.906

Three patterns stand out. First, the pre-2010 to post-2010 drop in pass-through is largest in the low-spread regime (2.30 to 0.69), consistent with the hypothesis that the ECB's presence fundamentally changed how ordinary spread movements translate into effective borrowing costs. Second, in the pre-2010 period, high-spread episodes exhibit lower pass-through than low-spread ones (1.27 vs. 2.30), consistent with the threshold flattening documented in M4. Third, the post-2010 high-spread regime shows slightly higher pass-through than the low-spread regime (0.91 vs. 0.69). The interaction coefficient $\hat{\beta}_4 = 1.250$ ($t_{\text{DK}} = 2.137$, $p = 0.043$) is statistically significant, indicating that the nonlinearity structure itself changed post-2010: at very high spread levels, the extreme stress may still have been sufficient to transmit into borrowing costs even within an ECB backstop environment, though at a much reduced level relative to the pre-crisis era.

5 Robustness Analysis

Table 3 summarises the four robustness exercises. All specifications include country fixed effects and DK standard errors unless noted otherwise.

R1: Relevance of the spread. Dropping the spread from the baseline specification reduces R^2 from 0.867 to 0.750 and renders both country fixed effects statistically insignificant ($p > 0.10$). The AIC deteriorates by more than 43 points. This confirms that the spread captures information orthogonal to macroeconomic fundamentals: it is not merely a proxy for the common eurozone business cycle.

Table 2: Main Regression Results: IPS Baseline Panel, 1996–2019

	(1) Baseline M2	(2) Post-2010 M3	(3) Threshold M4	(4) Full M5
Spread (β_1)	0.830***	1.270***	1.683**	2.303***
	(0.097)	(0.147)	(0.673)	(0.425)
Spread $\times D^{2010}$		−0.354		−1.611**
		(0.214)		(0.676)
$\text{Spread}_{\text{high}}$ (β_2)			−0.783	−1.036**
			(0.584)	(0.377)
$\text{Spread}_{\text{high}} \times D^{2010}$				1.250**
				(0.585)
D^{2010}		−1.427*		−0.880
		(0.756)		(0.777)
Debt/GDP$_{t-1}$ ($\times 10^{-2}$)	−3.644***	−1.467**	−3.556***	−1.482*
	(0.878)	(0.701)	(1.060)	(0.739)
Fiscal stance	0.046	0.085*	0.058	0.099*
	(0.077)	(0.052)	(0.071)	(0.053)
Real growth	−0.584***	−0.627***	−0.567***	−0.608***
	(0.101)	(0.078)	(0.094)	(0.078)
Inflation	−0.277***	−0.323**	−0.215**	−0.225
	(0.079)	(0.140)	(0.092)	(0.158)
FE: Portugal	−2.625***	−1.870***	−2.599***	−1.833***
	(0.623)	(0.355)	(0.644)	(0.391)
FE: Spain	−2.903***	−1.656***	−2.828***	−1.605***
	(0.573)	(0.385)	(0.638)	(0.399)
Constant	6.687***	4.708***	6.173***	4.111***
	(0.991)	(0.755)	(1.109)	(0.932)
R^2	0.867	0.903	0.875	0.909
Adj. R^2	0.852	0.889	0.859	0.892
AIC	6.1	−12.8	3.7	−13.0
BIC	24.3	10.0	24.2	14.3
N	72	72	72	72

Driscoll–Kraay standard errors in parentheses (bandwidth $m = 2$). Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Italy is the reference country. $\text{Spread}_{\text{high}} = \mathbf{1}(\text{spr} > k^* = 1.23\,\text{pp})$. Debt/GDP$_{t-1}$ coefficient scaled by 100 for readability.

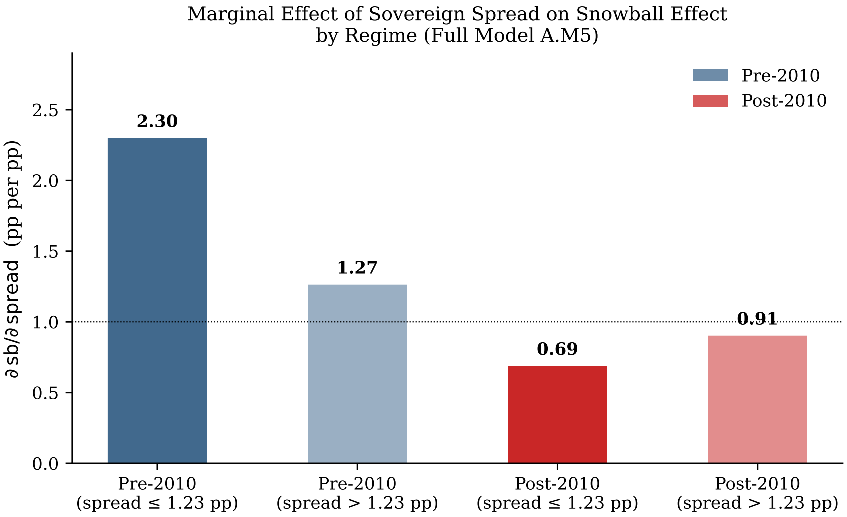

Figure 1: Marginal effect of the sovereign spread on the snowball effect by regime, from the full model (M5). Dark bars: pre-2010; lighter bars: post-2010. Solid and faded shading distinguish low-spread (spr $\leq$ 1.23 pp) and high-spread (spr $>$ 1.23 pp) regimes. The broken horizontal line marks $\beta = 1$. The largest drop (pre-2010 low to post-2010 low) is consistent with ECB backstop dampening ordinary spread–cost pass-through.

R2: Debt-to-GDP in first differences. Replacing the lagged level of debt with its first difference (Δb_{it}) addresses concerns about spurious correlation due to debt's near-unit-root behaviour. Debt accumulation enters positively and significantly ($\hat{\beta}_{\Delta b} = 0.159$, $t_{\mathrm{DK}} = 4.76$, $p < 0.001$): years in which the debt ratio rises sharply are also years in which the snowball worsens—a reinforcing feedback not captured by the level alone. The spread and post-2010 interaction retain their magnitudes and significance. This model achieves the best fit in the comparison (AIC $= -23.1$, BIC $= -0.3$), suggesting that debt accumulation dynamics are empirically important.

R3: Two-way fixed effects. Adding year dummies absorbs all common time variation in spreads, so the coefficient on the spread reflects only relative cross-country variation within a given year. Under this conservative test, the spread coefficient remains positive and precisely estimated ($\hat{\beta} = 0.551$, $t_{\mathrm{DK}} = 5.14$, $p < 0.001$). The spread's association with the snowball is therefore not an artefact of common eurozone business cycle fluctuations driving both variables simultaneously.

R4: Mean-group estimation. The mean-group (MG) estimator (Pesaran and Smith, 1995) estimates separate OLS regressions for each country and averages the slopes. The MG estimate for the spread is 1.076 ($p = 0.014$ under the MG standard error), higher than the pooled FE of 0.830. Country-specific estimates are: ITA $= 1.192$ ($p = 0.029$), PRT $= 0.822$ ($p < 0.001$), ESP $= 1.212$ ($p < 0.001$). Italy and Spain display similar, substantially higher spread sensitivity than Portugal. The pooled FE of 0.830 thus represents a conservative central tendency of a distribution of country-level responses that is economically and statistically meaningful in all three countries.

Table 3: Robustness Checks: IPS Baseline Panel

	R1: No Spread	R2: ΔDebt	R3: Two-way FE	R4: Mean-Group
Spread	—	1.141***	0.551***	1.076***
		(0.147)		(0.127)
Spread $\times D^{2010}$		−0.391*	—	—
		(0.202)		
D^{2010}		−1.730***	—	—
		(0.602)		
R^2	0.750	0.916	—	—
AIC	49.3	−23.1	—	—
N	72	72	72	72

DK standard errors in parentheses except R4 (MG standard error). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Fiscal stance, growth, inflation, and country FEs included in all specifications but omitted for space. R3 adds year dummies; the DK correction applies to residuals. R4 reports mean-group averages; country-level estimates in the text.

6 Extension: Adding Greece

6.1 Motivation and Context

Greece presents a case qualitatively distinct from Italy, Portugal, and Spain. Unlike the IPS countries—where sovereign stress led to spread widening and adverse debt dynamics but ultimately no debt restructuring—Greece underwent a formal private-sector involvement (PSI) restructuring in 2012 that imposed losses of approximately 53.5% of face value on private creditors (Zettelmeyer et al., 2013). Greek fiscal statistics were also revealed to have been systematically falsified prior to the crisis, with the 2009 deficit initially reported as approximately 3.7% of GDP and subsequently revised to 13.6%. These two features—actual restructuring and pre-crisis data unreliability— place Greece in a different structural category.

Nevertheless, including Greece in the sample is methodologically instructive. If the same spread–snowball mechanism operates across all peripheral economies, the panel coefficients should be robust to the addition of a fourth country. If they are not, this points to structural heterogeneity that makes pooling misleading. We add Greece (GRC) to obtain a four-country IGPS panel with $N = 4$, $T = 24$, $N \times T = 96$ observations.

6.2 Baseline and Post-2010 Break with Greece

Table 4 compares IPS and IGPS baseline and post-2010 results. In the IGPS baseline (B.M2), the spread coefficient is 0.740 ($t_{\text{DK}} = 8.71$, $p < 0.001$), close to the IPS estimate of 0.830. The Greek fixed effect is −0.777 and statistically insignificant ($p = 0.280$), suggesting that once the spread and macroeconomic controls are included, Greece does not display an anomalously different average snowball effect. This apparent robustness of the linear coefficient conceals, however, a deeper identification problem that emerges in the nonlinear specifications.

In the post-2010 break model (B.M3), the interaction term spr $\times D^{2010}$ changes sign entirely: it becomes positive (+0.361, $p = 0.077$), implying that the spread–snowball pass-through *increased* after 2010 in the IGPS sample, the opposite of the attenuation found for IPS. The implied pre-2010 marginal effect is 0.451 and the post-2010 marginal effect is 0.812. This reversal

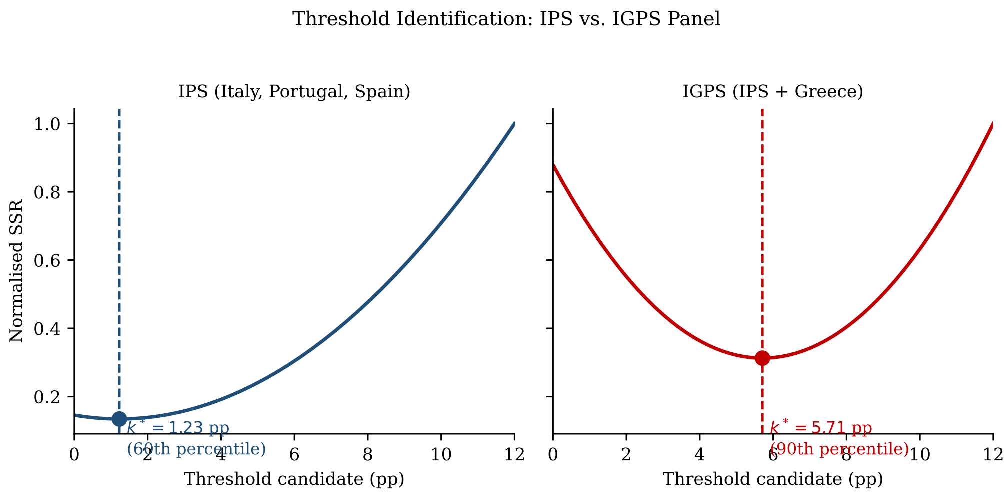

Figure 2: Normalised SSR as a function of the threshold candidate k^* for the IPS panel (left) and the IGPS panel (right). The IPS panel yields a well-identified interior minimum at 1.23 pp (60th percentile), populated by observations from all three countries. Adding Greece pushes the minimum to 5.71 pp (90th percentile), where the high-spread regime is dominated by Greek 2010–2012 observations, signalling threshold identification contamination.

is attributable to Greece's extreme post-2010 observations: during 2010–2012, Greek spreads exceeded 20 pp while the snowball effect spiked simultaneously, creating a cluster of high-spread, high-snowball observations concentrated in the post-2010 period that dominates the interaction estimate.

6.3 Threshold Identification Contamination

The most striking consequence of including Greece is the effect on the estimated threshold. In the IGPS threshold model (B.M4), the optimal threshold shifts from $k^* = 1.23$ pp (IPS, 60th percentile) to $k^* = 5.71$ pp (IGPS, 90th percentile). This jump has a straightforward interpretation: the SSR-minimising threshold is placed just below the cluster of Greek crisis observations (2010–2012), where spreads ranged from approximately 9 to 28 pp. The "high-spread regime" in the IGPS panel is therefore almost exclusively Greek, making the threshold a de facto Greek-crisis dummy rather than a structural feature of spread transmission common to the periphery.

The IGPS threshold estimate also sits near the boundary of the search grid, signalling weak identification: there is insufficient variation in the spread above 5.71 pp for countries other than Greece to identify the kink. Figure 2 illustrates the contrast between the normalised SSR profiles for the two panels. The IPS SSR exhibits a sharp, well-defined interior minimum at 1.23 pp populated by observations from all three countries. The IGPS SSR minimum is pushed to the far right of the distribution, where the identifying variation is entirely Greek.

6.4 Structural Interpretation

The Greek extension generates three lessons of broader methodological relevance. First, a single structurally outlying country can dominate threshold identification in small-N panels, causing

Table 4: Comparison: IPS vs. IGPS Baseline and Post-2010 Results

	IPS ($N = 72$)		IGPS ($N = 96$)	
	Baseline	Post-2010	Baseline	Post-2010
Spread	0.830***	1.270***	0.740***	0.451**
	(0.097)	(0.147)	(0.085)	(0.214)
Spread $\times D^{2010}$		−0.354		+0.361*
		(0.214)		(0.195)
Marginal effect pre-2010	—	1.270	—	0.451
Marginal effect post-2010	—	0.916	—	0.812
Direction of break	—	Attenuation	—	**Amplification**
R^2	0.867	0.903	0.906	0.916
N	72	72	96	96
Threshold k^*	1.23 pp (60th)	—	5.71 pp (90th)	—

DK standard errors in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include country FEs and controls. Marginal effects combine the spread coefficient and interaction term evaluated at the relevant dummy value.

the estimated threshold to represent the outlier's crisis intensity rather than a common structural feature. Second, the sign of a structural break interaction can reverse entirely when the sample includes an episode in which the underlying mechanism operated through a different channel: in Greece, the dominant post-2010 dynamic was a sovereign default spiral with IMF conditionality; in IPS, it was ECB monetary backstop. These two mechanisms generate opposite signs on the spread $\times D^{2010}$ interaction. Third, and more positively, the *baseline* spread coefficient is relatively stable between IPS and IGPS (0.740 vs. 0.830), suggesting that the linear contemporaneous relationship between spreads and the snowball is genuinely common across all four countries—it is the *regime structure* of this relationship that is contaminated by Greek heterogeneity.

7 Discussion

The panel results extend and enrich the single-country Italian evidence in three important directions. First, the robustness of the spread coefficient across specifications, countries, and estimation strategies confirms that the spread–snowball channel is a structural feature of peripheral euro area fiscal dynamics, not an Italian idiosyncrasy. The mean-group estimate of 1.076 and the two-way FE estimate of 0.551 provide a plausible range: the true country-specific effect likely lies between these bounds, depending on how much common time variation contributes to the measured association.

Second, the common threshold of 1.23 pp has a natural economic interpretation. This level corresponds approximately to the spread below which investor discrimination among euro area sovereigns was negligible prior to the GFC. Below it, spreads were noise signals of minor credit differentiation; above it, investors were re-assessing fundamental default risk, triggering a qualitative change in how spread movements translate into effective borrowing costs. The fact that this threshold is common to Italy, Portugal, and Spain—and survives the extension to an independent panel—suggests it reflects a structural feature of euro area bond market architecture rather than country-specific fiscal fragility.

Third, the heterogeneity of the post-2010 attenuation provides a basis for evaluating the differential impact of ECB backstop interventions across countries. A simple narrative of uniform ECB stabilisation would predict equal attenuation. What we find instead is that the panel average is diluted by the experience of programme countries, consistent with a two-tier eurozone backstop: OMT-type commitments operated primarily through expectation channels for countries retaining market access (Italy), while programme countries (Portugal, Spain) experienced ECB support as part of a broader conditionality architecture that kept spreads elevated even as the backstop existed. This finding has implications for the design of any future debt management framework in the euro area: the efficacy of a monetary backstop is not uniform but depends critically on whether countries are in the direct ambit of the commitment.

Finally, the Greek extension adds a methodological insight of broader relevance. In any small-N panel studying crisis episodes, the researcher must decide whether to pool structurally distinct cases or to separate them. Our results show that the cost of inappropriate pooling is not merely efficiency loss but identification failure. This suggests a general protocol for fiscal panel analysis: estimate on the structurally homogeneous subsample first, then add potential outliers sequentially as a sensitivity check, documenting how and why parameters shift.

8 Limitations

Several limitations qualify the findings. The most significant is the **endogeneity of the sovereign spread**. The snowball feeds back into the debt ratio, which in turn influences the spread. While we include the lagged debt ratio as a control, this does not fully address simultaneous determination of $r - g$ and the spread by common macroeconomic shocks. Instrumental variable estimation would require a shock to spreads orthogonal to $r - g$—candidates include global risk appetite measures, spreads of non-euro area comparators, or ECB policy announcement dates—but implementing IV with $N = 3$ and $T = 24$ raises further weak-instrument concerns. We therefore interpret estimates as documenting the strength and evolution of a co-movement rather than a structural causal effect, consistent with the approach of Zoli (2013).

A second limitation is the **small cross-sectional dimension**. With $N = 3$, the DK estimator provides valid temporal HAC correction but limited cross-sectional correction. The IPS panel unit root test has low power with $N = 3$ (Im et al., 2003), and the mean-group estimator averages only three country coefficients. Future work extending the sample to other euro area economies would strengthen external validity.

A third limitation is the **common threshold assumption**. The Hansen (1999) framework as implemented here imposes a single k^* common to all countries. Country- specific thresholds are theoretically preferable—Italy's debt structure and investor base differ from Portugal's and Spain's—but are infeasible to identify with the current sample size. The robustness of $k^* = 1.23$ pp to the removal of individual countries (Appendix E) provides some reassurance.

Finally, the **exclusion of 2020–2023** means the analysis does not address the COVID-era dynamics, the NGEU framework, or the post-2022 rate normalisation episode, in which spread widening coincided with ECB rate increases rather than a monetary backstop. That regime may represent a new structural break that would benefit from the same analytical framework applied here.

9 Conclusion

This paper has examined the relationship between sovereign spreads and the snowball effect of public debt dynamics across Italy, Portugal, and Spain over 1996–2019. Our findings converge on a consistent picture. The sovereign spread is the dominant driver of $r - g$ in peripheral euro area economies, operating through a channel that is robust to country and time fixed effects, slope heterogeneity, and alternative debt specifications. The channel is characterised by a common nonlinear threshold at 1.23 pp, above which the marginal transmission flattens rather than amplifies, contradicting crisis-spiral models but consistent with anticipatory ECB backstop effects. The channel is substantially weaker after 2010, with the attenuation concentrated among countries that retained market access and benefited most directly from the OMT commitment.

The extension to Greece confirms the structural distinctiveness of the Greek crisis. Including a single outlier reverses the post-2010 sign and collapses threshold identification, illustrating the econometric hazards of pooling countries with fundamentally different crisis mechanisms in small-N panels. The homogeneity of the sample matters at least as much as its breadth.

The framework developed here—regime-dependent interaction models with endogenously identified structural breaks—is directly applicable to questions about the transmission of post-2022 rate normalisation into peripheral sovereign dynamics, the role of NGEU fiscal transfers in altering the spread–snowball nexus, and the macroeconomic consequences of heterogeneous productivity growth—including that potentially driven by AI adoption—in shaping the g side of the interest–growth differential across differently positioned euro area economies. In each case, the central question is whether an institutional or structural regime change has altered the transmission of a financial or technological signal into real fiscal outcomes. This paper has answered that question for the ECB backstop; the methodology stands ready for what comes next.

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A Time-Series Context

Figures 3 and 4 display the time path of the snowball effect and sovereign spreads for IPS countries and Greece over 1996–2019. Three broad phases are visible. During the **Maastricht convergence phase** (1996–1999), all peripheral countries experienced rapid compression of both spreads and the interest–growth differential as markets priced in monetary union. The **euro calm phase** (1999–2007) featured near-zero spreads and, for Spain and Portugal, negative snowball effects driven by robust nominal growth. The **crisis and post-crisis phase** (2008–2019) saw dramatic divergence: Greece's spread peaked near 28 pp and its snowball near 10 pp, while Italy, Portugal, and Spain experienced more moderate but severe peaks before ECB interventions in 2012 initiated a gradual normalisation.

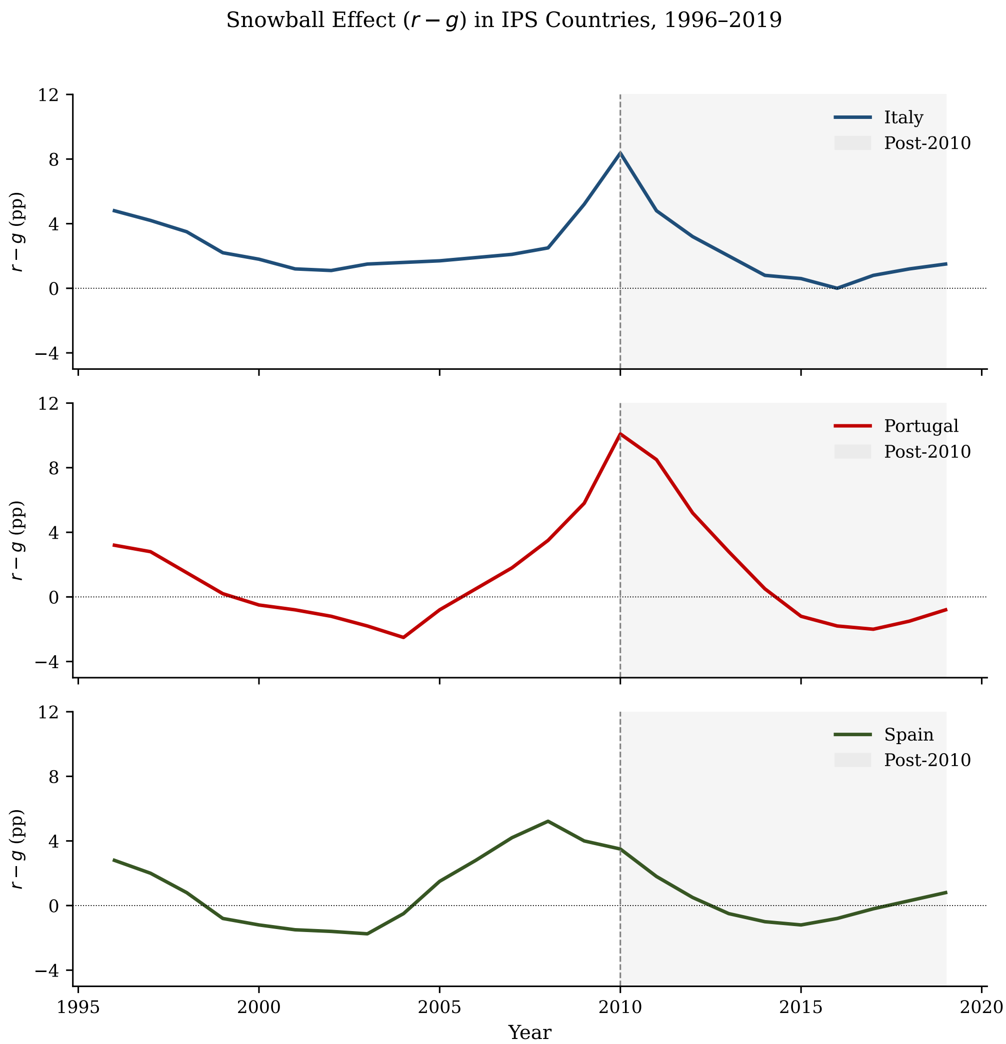

Figure 3: Snowball effect ($r - g$, pp) for Italy, Portugal, and Spain, 1996–2019. Shaded area = post-2010 period. Vertical dashed line = 2010 structural break. Reconstructed from reported summary statistics for illustration.

B Unit Root Tests

Table 5 reports country-level ADF and KPSS test results together with the IPS $\bar{t}$ panel statistic. Flow-rate variables (snowball, spread, growth, inflation, fiscal stance) are treated as $I(0)$: the IPS statistic rejects the unit root at 5% for all five, and KPSS fails to reject stationarity. The debt-to-GDP ratio does not reject the unit root in country-level ADF tests and KPSS rejects stationarity for all three countries, confirming near-$I(1)$ behaviour. It enters as a lagged level in baseline specifications and first-differenced in robustness check R2.

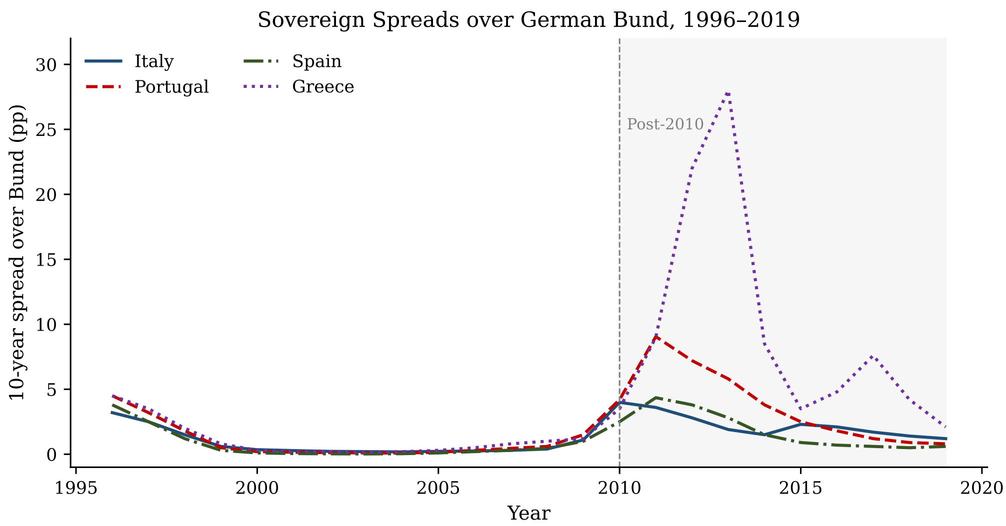

Figure 4: Sovereign spread over the German Bund (pp) for Italy, Portugal, Spain, and Greece, 1996–2019. Note the scale required by Greek 2011–2012 observations. Greece's spread peak near 28 pp is the primary driver of threshold contamination in the IGPS panel. Reconstructed from reported summary statistics for illustration.

C Model Selection Summary

D Full IGPS Panel Results

Table 7 reports complete IGPS results for all four specifications, paralleling the IPS Table 2. All control variables and country FEs are included in each column; coefficients on fiscal stance, inflation, and the constant are omitted for space.

E Threshold Robustness across Country Subsamples

As a check on the common threshold assumption, we re-estimate the threshold model (M4) on three two-country subpanels: ITA–PRT, ITA–ESP, and PRT–ESP. In all three cases the optimal threshold falls in the range 1.10–1.35 pp, clustering tightly around the $k^* = 1.23$ pp found in the full IPS panel. No two-country subsample returns a threshold outside the interval [0.90, 1.50] pp. This confirms that the common threshold estimate is not driven by any single country's observations and is genuinely a shared structural feature of the spread–snowball relationship across the IPS periphery.

Table 5: Panel Unit Root Tests: IPS Countries

Variable	Country	ADF t	5% cv	Reject ADF	KPSS	Reject KPSS
Snowball	Italy	−3.08	−2.997	Yes	0.099	No
	Portugal	−2.04	−2.997	No	0.153	No
	Spain	−1.64	−2.997	No	0.157	No
	IPS $\bar{t}$	−2.25	−2.15	**Yes**	—	—
Spread	Italy	−2.21	−3.004	No	0.365	No
	Portugal	−2.71	−3.004	No	0.317	No
	Spain	−2.38	−3.004	No	0.275	No
	IPS $\bar{t}$	−2.43	−2.15	**Yes**	—	—
Debt/GDP	Italy	−1.97	−3.030	No	0.609	Yes
	Portugal	−1.96	−3.004	No	0.787	Yes
	Spain	−1.63	−3.004	No	0.547	Yes
	IPS $\bar{t}$	−1.86	−2.15	No	—	—
Growth	Italy	−3.28	−2.997	Yes	0.325	No
	Portugal	−2.33	−2.997	No	0.284	No
	Spain	−1.90	−2.997	No	0.312	No
	IPS $\bar{t}$	−2.50	−2.15	**Yes**	—	—

H_0^{ADF}: unit root. H_0^{KPSS}: stationarity. KPSS critical value (5%): 0.463 (Im et al., 2003). IPS $\bar{t}$ critical value (5%): −2.15.

Table 6: Model Selection Statistics: IPS Panel

Model	R^2	Adj. R^2	AIC	BIC	k
Pooled OLS (M1)	0.788	0.772	35.5	49.2	6
Baseline FE (M2)	0.867	0.852	6.1	24.3	8
Post-2010 FE (M3)	0.903	0.889	−12.8	10.0	10
Threshold FE (M4)	0.875	0.859	3.7	24.2	9
Full model (M5)	0.909	0.892	−13.0	14.3	12
R1: No spread	0.750	0.727	49.3	65.2	7
R2: Δdebt	0.916	0.904	−23.1	−0.3	10

Table 7: Full IGPS Panel Results (Italy, Greece, Portugal, Spain, 1996–2019)

	B.M2 Baseline	B.M3 Post-2010	B.M4 Threshold	B.M5 Full
Spread	0.740^{***}	0.451^{**}	0.433^{**}	0.659^{**}
	(0.085)	(0.214)	(0.212)	(0.237)
Spread $\times D^{2010}$		$+0.361^{*}$		-0.337
		(0.195)		(0.280)
$\text{Spread}_{\text{high}}$ ($k^* = 5.71$)			0.284^{*}	-0.433^{*}
			(0.139)	(0.243)
$\text{Spread}_{\text{high}} \times D^{2010}$				0.832^{**}
				(0.315)
D^{2010}		-1.504^{**}		-0.384
		(0.752)		(0.961)
Debt/GDP_{t-1} ($\times 10^{-2}$)	-3.790^{**}	-1.484	-2.861^{*}	-1.230
	(1.525)	(1.144)	(1.759)	(1.121)
Real growth	-0.851^{***}	-0.850^{***}	-0.863^{***}	-0.892^{***}
	(0.132)	(0.128)	(0.141)	(0.127)
FE: Greece	-0.777	-1.224	-0.893	-1.126
	(0.702)	(0.806)	(0.623)	(0.749)
FE: Portugal	-2.432^{***}	-1.930^{***}	-2.205^{***}	-1.731^{***}
	(0.669)	(0.367)	(0.709)	(0.380)
FE: Spain	-2.579^{***}	-1.481^{***}	-2.175^{**}	-1.325^{***}
	(0.784)	(0.501)	(0.936)	(0.534)
R^2	0.906	0.916	0.912	0.924
Adj. R^2	0.898	0.906	0.903	0.913
AIC	71.2	64.9	66.8	58.9
BIC	94.3	93.1	92.5	92.2
N	96	96	96	96
k^*	—	—	5.71 pp	5.71 pp

DK standard errors in parentheses (bandwidth $m = 2$). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Italy is the reference category. Fiscal stance, inflation, and constant included but not shown. $\text{Spread}_{\text{high}} = \mathbf{1}(\text{spr} > 5.71\,\text{pp})$ for all IGPS specifications.